

ADMISSION OPENS FOR BTech IN ARTIFICIAL INTELLIGENCE

Welcome to BTech AI!

Embark on a transformative journey with our BTech in Artificial Intelligence program. Designed to equip you with cutting-edge skills in AI, machine learning, and data science, this program opens doors to a future of innovation and opportunity.



1. Introduction

The Bachelor of Technology (BTech) in Artificial Intelligence (AI) is an undergraduate engineering program designed to equip students with a strong foundation in computer science, mathematics, and core AI methodologies. It focuses on developing intelligent systems capable of simulating human decision-making, learning, and problem-solving. The program integrates both theoretical understanding and practical skills required to design and deploy AI-based applications across diverse sectors.

2. Program Objectives

- To impart core knowledge of artificial intelligence, machine learning, deep learning, and data science.
- To build strong foundations in mathematics, programming, and algorithmic thinking.
- To expose students to real-world applications of AI in sectors such as healthcare, agriculture, education, finance, robotics, natural language processing, and computer vision.
- To promote innovation and entrepreneurship through project-based learning and industry collaboration.
- To prepare students for careers in AI research, development, and deployment, both in Nepal and globally.

3. Core Course Areas

The BTech in Artificial Intelligence program is spread over 8 semesters and combines core engineering subjects with specialized AI courses. The curriculum is designed to build a strong technical foundation while also introducing students to the latest developments in the AI field.

Key Subjects Covered:

- **Programming and Software Development**
Students learn essential programming languages and techniques such as C, C++, Python, Object-Oriented Programming (OOP), Data Structures, Algorithms, Design and Analysis of Algorithms.
- **Mathematics for AI**
Core mathematical subjects such as Calculus, Linear Algebra, Probability, Statistics, Numerical Methods, and Differential Equations are taught to strengthen analytical and problem-solving skills.
- **Machine Learning and Data Science**
Courses in Machine Learning, Advanced Machine Learning, Data Science, and Data Mining help students understand how to train intelligent models and extract insights from large datasets.
- **Computer Vision and Natural Language Processing (NLP)**
Students are introduced to AI fields that deal with images and language through courses like Pattern Recognition, Image Processing, Language Models, Deep Learning, and Information Retrieval.
- **Additional Technology Courses**
 - Computer Networks and Cybersecurity – to understand data communication and secure system design.

- Embedded Systems – for learning how AI can be implemented in hardware devices.
- Internet of Things (IoT) and Blockchain – to explore modern technologies that integrate well with AI in areas like smart systems and handling.
- Fuzzy Optimization – a method used to handle uncertainties in optimization problems, emerging as a new area within operations research mathematics, with special applications in AI.

4. Industry Preparation and Electives

From 6th to 8th semesters, students can also choose from a wide range of elective subjects. These electives are designed to address current and emerging trends in the AI industry, allowing students to specialize based on their interests and career goals.

In the 7th semester, students take a dedicated Industry Preparation course that equips them with professional and soft skills needed for jobs.

In the 8th semester, students are required to complete a mandatory internship, giving them valuable hands-on experience in real industry support this, the curriculum in the final semester is kept flexible. We have also signed Memorandums of Understanding (MoUs) with software companies in Nepal—such as Fusemachines, Leapfrog, and others—to help students secure meaningful internship opportunities.

5. Relevance of AI in Today's World

Artificial Intelligence has emerged as a transformative force shaping the future of technology. From autonomous vehicles and voice assistants to agriculture and predictive analytics, AI is at the forefront of innovation. The exponential growth of data, advancements in computing power in machine learning algorithms have expanded the boundaries of what AI can achieve. In a developing country like Nepal, AI holds great potential for addressing local challenges through customized solutions in areas such as:

- Smart healthcare systems for diagnosis and treatment planning
- AI-powered agriculture tools for yield prediction and pest detection
- Natural language interfaces for local languages and dialects
- Predictive analytics for disaster management and governance

6. Scope of AI in Nepal

Nepal is gradually embracing AI technologies, supported by a growing ecosystem of startups, academic institutions, and government initiatives. As the industry is still in its early development, there is promising scope in several core areas:

- **E-Governance:** Automating administrative processes, decision-support systems, digital document analysis, and citizen service automation.
- **Education Technology:** Development of adaptive learning systems, intelligent tutoring, automated evaluation tools, and educational content creation.
- **Agriculture:** AI applications for crop health monitoring, soil analysis, smart irrigation systems, and market demand forecasting.
- **Natural Language Processing (NLP):** Tools for Nepali-English translation, voice assistants, sentiment analysis, and chatbot development.
- **Computer Vision:** Facial recognition, surveillance systems, quality inspection in manufacturing, and image-based data analysis.
- **Software Automation:** Intelligent task scheduling, predictive analytics, and workflow optimization for businesses and enterprises.

Several companies are actively working in the AI and software domain in Nepal. Some notable names include:

- Fusemachines Nepal – AI education, natural language processing, and AI-powered solutions.
- Leapfrog Technology – AI integration in healthcare and enterprise software.
- Cotiviti Nepal – Data analytics and machine learning in large-scale systems.
- LogPoint – Security analytics using machine learning.
- CloudFactory – Human-in-the-loop AI services and data labeling.
- GrowByData – E-commerce data analytics and AI tools.
- Neema Academy – AI-based education technology.
- Rumsan – Blockchain and data-driven digital solutions.
- Ingrails – AI integration in school management and enterprise systems.

7. Career Opportunities

Graduates of the BTech in AI program are well-positioned for a variety of roles in both local and international job markets. Career opportunities include:

- AI Engineer
- Data Scientist
- Machine Learning Engineer
- Computer Vision Engineer
- NLP Researcher

- Robotics Engineer
- Business Intelligence Analyst
- AI Product Manager
- AI Consultant
- Research Associate (AI and Data Science)

In addition to industrial jobs, academic paths, students are qualified to apply for officer-level positions in the government sector by appearing in the National Service Commission (Lok Sewa Aayog) examinations.

8. Further Education

The Department of Artificial Intelligence also offers two specialized postgraduate programs:

1. **MTech in AI:** Designed for students who have completed a four-year bachelor's degree in fields such as Computer Science, Information Technology, Electronics, or Software Engineering.
2. **Master's in AI Application:** Open to students from any academic background who have completed a four-year bachelor's degree (and a bachelor's degree with one year bridge course). This program focuses on practical applications of AI across various domains.

In addition to these, students can pursue further studies in related fields such as Computer Science, Data Science, or Management—both domestically and at renowned universities abroad.

More info: <https://soe.ku.edu.np/admission>

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